

## MATTMAR V, Sweden

WMO: 022160

Lat: 63.3054N

Lon: 13.8767E

Elev: 360

StdP: 97.07

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -23.8      | -20.4 | -25.7                         | 0.4 | -23.8 | -22.3 | 0.5 | -20.2 | 11.1                  | 1.8  | 9.9 | 0.1  | 0.3                   | 320  | 0.690 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.6                    | 24.6            | 16.6 | 22.6 | 15.7 | 20.5 | 14.7 | 17.5                | 22.7 | 16.5 | 21.2 | 15.4 | 19.5 | 3.2                  | 140  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 15.4                            | 11.5 | 19.5 | 14.4 | 10.7 | 18.3 | 13.5 | 10.1 | 17.4 | 50.5          | 22.5 | 47.4 | 21.3 | 44.3 | 19.6 | 20.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| (4) 9.7           | 8.4  | 7.3 | -26.7                      | 27.1 | 3.6                | 1.9 | -29.3                                              | 28.5 | -31.4      | 29.7 | -33.4      | 30.8 | -36.0      | 32.2 |  |  |
| (5)               |      |     | -26.7                      | 18.8 | 3.4                | 1.4 | -29.1                                              | 19.8 | -31.1      | 20.7 | -33.0      | 21.5 | -35.4      | 22.5 |  |  |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 3.2   | -6.9  | -6.0  | -2.4  | 2.8   | 7.2   | 11.6  | 14.5  | 12.9  | 8.5   | 3.1   | -1.6  | -5.4 | (6)  |
| (7)  |                                                                                  | DBStd       | 8.68  | 6.56  | 6.48  | 4.71  | 3.36  | 3.60  | 3.65  | 3.14  | 2.92  | 3.06  | 3.86  | 4.62  | 6.62 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 2809  | 525   | 448   | 385   | 217   | 103   | 22    | 2     | 7     | 62    | 214   | 347   | 477  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 5515  | 783   | 681   | 643   | 467   | 345   | 203   | 124   | 170   | 294   | 471   | 597   | 736  | (9)  |
| (10) |                                                                                  | CDD10.0     | 347   | 0     | 0     | 0     | 1     | 17    | 71    | 142   | 97    | 18    | 1     | 0     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 10    | 0     | 0     | 0     | 0     | 0     | 2     | 7     | 2     | 0     | 0     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 107   | 0     | 0     | 0     | 0     | 3     | 31    | 59    | 13    | 0     | 0     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 7     | 0     | 0     | 0     | 0     | 0     | 1     | 6     | 1     | 0     | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 2.9   | 2.6   | 2.8   | 3.3   | 3.2   | 3.4   | 3.4   | 2.9   | 2.6   | 2.9   | 3.0   | 2.6   | 2.5  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 648   | 44    | 35    | 35    | 35    | 44    | 70    | 93    | 83    | 56    | 55    | 50    | 43   | (15) |
| (16) |                                                                                  | PrecMax     | 912   | 74    | 77    | 60    | 68    | 109   | 127   | 199   | 151   | 115   | 170   | 103   | 96   | (16) |
| (17) |                                                                                  | PrecMin     | 481   | 4     | 4     | 8     | 7     | 12    | 17    | 20    | 35    | 17    | 14    | 11    | 18   | (17) |
| (18) |                                                                                  | PrecStd     | 114   | 18    | 16    | 13    | 17    | 24    | 31    | 50    | 37    | 27    | 34    | 21    | 15   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 6.0   | 5.9   | 8.7   | 16.6  | 23.2  | 25.9  | 27.5  | 25.1  | 19.1  | 13.5  | 10.6  | 6.5  | (19) |
| (20) |                                                                                  |             | MCWB  | 3.5   | 3.2   | 4.3   | 10.1  | 15.6  | 16.8  | 18.5  | 16.7  | 13.5  | 9.4   | 7.7   | 4.5  | (20) |
| (21) |                                                                                  | 2%          | DB    | 4.3   | 4.4   | 6.5   | 13.1  | 19.8  | 24.2  | 25.0  | 22.3  | 16.7  | 11.4  | 7.9   | 5.2  | (21) |
| (22) |                                                                                  |             | MCWB  | 2.4   | 2.5   | 3.5   | 7.6   | 12.8  | 16.2  | 17.1  | 16.1  | 12.2  | 8.2   | 5.9   | 3.3  | (22) |
| (23) |                                                                                  | 5%          | DB    | 2.7   | 3.3   | 5.1   | 10.7  | 16.6  | 21.3  | 23.0  | 20.4  | 15.1  | 10.2  | 5.7   | 4.0  | (23) |
| (24) |                                                                                  |             | MCWB  | 1.3   | 1.6   | 2.8   | 6.2   | 11.0  | 14.7  | 16.0  | 15.2  | 11.7  | 7.7   | 4.0   | 2.4  | (24) |
| (25) |                                                                                  | 10%         | DB    | 1.4   | 2.0   | 3.9   | 8.6   | 13.7  | 18.7  | 21.0  | 18.5  | 13.5  | 8.9   | 4.1   | 2.7  | (25) |
| (26) |                                                                                  |             | MCWB  | 0.2   | 0.5   | 1.9   | 5.0   | 9.3   | 13.4  | 15.1  | 14.5  | 10.5  | 7.1   | 2.7   | 1.1  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 3.8   | 3.6   | 5.3   | 10.4  | 15.9  | 17.7  | 19.1  | 18.2  | 14.5  | 10.7  | 8.0   | 4.8  | (27) |
| (28) |                                                                                  |             | MCDWB | 5.5   | 5.2   | 7.8   | 16.1  | 23.4  | 24.1  | 26.0  | 22.6  | 17.5  | 12.1  | 10.1  | 6.3  | (28) |
| (29) |                                                                                  | 2%          | WB    | 2.6   | 2.7   | 4.0   | 8.1   | 13.3  | 16.5  | 17.9  | 17.0  | 13.0  | 9.0   | 6.1   | 3.5  | (29) |
| (30) |                                                                                  |             | MCDWB | 4.0   | 4.2   | 6.1   | 12.5  | 18.5  | 22.9  | 23.7  | 20.6  | 15.7  | 10.8  | 7.8   | 5.0  | (30) |
| (31) |                                                                                  | 5%          | WB    | 1.5   | 1.7   | 3.0   | 6.7   | 11.6  | 15.2  | 16.8  | 15.8  | 11.9  | 8.1   | 4.3   | 2.5  | (31) |
| (32) |                                                                                  |             | MCDWB | 2.7   | 3.0   | 4.9   | 9.9   | 15.7  | 20.7  | 21.7  | 19.2  | 14.4  | 9.7   | 5.5   | 3.8  | (32) |
| (33) |                                                                                  | 10%         | WB    | 0.1   | 0.5   | 2.0   | 5.4   | 9.8   | 13.7  | 15.7  | 14.9  | 10.9  | 7.0   | 2.8   | 1.3  | (33) |
| (34) |                                                                                  |             | MCDWB | 1.2   | 1.8   | 3.7   | 8.1   | 13.6  | 18.1  | 19.8  | 18.0  | 13.1  | 8.9   | 3.9   | 2.5  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 5.9   | 6.5   | 7.4   | 8.1   | 8.6   | 8.7   | 8.6   | 8.1   | 7.0   | 5.7   | 4.9   | 5.6  | (35) |
| (36) |                                                                                  |             | MCDBR | 6.0   | 6.1   | 7.4   | 12.1  | 13.3  | 13.2  | 12.1  | 11.3  | 9.7   | 7.6   | 6.9   | 6.2  | (36) |
| (37) |                                                                                  |             | MCWBR | 5.1   | 5.0   | 5.2   | 7.3   | 7.5   | 6.6   | 5.6   | 6.0   | 5.8   | 5.3   | 5.5   | 5.0  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 6.3   | 6.0   | 6.8   | 11.0  | 12.4  | 12.4  | 11.4  | 9.7   | 8.4   | 7.0   | 6.9   | 5.8  | (38) |
| (39) |                                                                                  |             | MCWBR | 5.4   | 5.1   | 5.0   | 6.9   | 7.3   | 6.5   | 5.7   | 5.7   | 5.5   | 5.3   | 5.6   | 4.9  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.200 | 0.262 | 0.292 | 0.324 | 0.332 | 0.329 | 0.342 | 0.331 | 0.303 | 0.286 | 0.211 | 0.185 | (40) |      |
| (41) |                                                                                  | taud        | 2.035 | 2.245 | 2.307 | 2.397 | 2.477 | 2.515 | 2.491 | 2.512 | 2.574 | 2.480 | 2.112 | 1.988 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 443   | 689   | 813   | 861   | 886   | 895   | 870   | 850   | 812   | 671   | 421   | 233   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 29    | 57    | 81    | 93    | 94    | 92    | 92    | 83    | 64    | 47    | 26    | 14    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.17  | 0.73  | 2.10  | 3.87  | 4.63  | 5.06  | 4.79  | 3.62  | 2.16  | 0.93  | 0.26  | 0.08  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.02  | 0.10  | 0.22  | 0.31  | 0.38  | 0.42  | 0.54  | 0.33  | 0.21  | 0.12  | 0.04  | 0.01  | (45) |      |

## Historical Trends

|                              |       | Heating |        | Cooling |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP       | HDD10.0 | HDD18.3 | CDD18.3 |
| (46) Station Only            | DBAvg | (d)     | (e)    | (f)     | (g)   | (h)         | (i)     | (j)     | (m)     |
| (47) Regional (46 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page